Quantum Mechanics II — Exercise Sheet 2

Evolution and a position reading

Variant: Solutions

Exercise 1 (A state with a shorter revival). Let $\psi = \sum_{j\geq 0} c_{2j+1} u_{2j+1}$ be normalized. (i) Show that its physical state revives at $T_{\mathrm{rev}}/8$, where $T_{\mathrm{rev}} = 2\pi\hbar/E_1$. (ii) Decide whether this time is always the least positive revival time. For a state with both c_1 and c_3 nonzero and all other coefficients zero, find that least time.

Solution. The odd squares satisfy $(2j+1)^2 = 1 + 8j(j+1)/2$. At $t = T_{\mathrm{rev}}/8$ every occupied mode has the same phase:

$$e^{-\mathrm{i}E_{2j+1}t/\hbar} = e^{-\mathrm{i}\pi/4}.$$

Thus the vector returns up to a common phase. This need not be the least time: a single energy eigenstate is stationary as a physical state. When only modes 1 and 3 are occupied, their relative phase returns exactly when $8E_1 t/\hbar \in 2\pi\mathbb{Z}$. The least positive time is $T_{\mathrm{rev}}/8$.

Exercise 2 (A position filter at a node). Start with the box mode u_3 and retain a position reading in $(0, L/3)$. (i) Find its probability and conditional vector. (ii) Compute all sine coefficients of the conditional vector. (iii) Decide whether its mean energy is finite and whether it belongs to $D(H)$. (iv) Treat the coefficient at $n = 3$ separately.

Solution. The integral of $|u_3|^2$ over one of its three equal lobes is $1/3$. Thus the probability and conditional vector are

$$p = \frac{1}{3}, \qquad \phi = \sqrt{3}\,\mathbf{1}_{(0,L/3)} u_3.$$

The coefficient is $2\sqrt{3}\int_0^{1/3} \sin(3\pi y)\sin(n\pi y)\,\mathrm{d}y$. The product-to-sum identity gives

$$c_3 = \frac{1}{\sqrt{3}}, \qquad c_n = \frac{6\sqrt{3}\sin(n\pi/3)}{\pi(9-n^2)}, \quad n \neq 3.$$

The coefficients have order n^{-2}, so $\sum n^2|c_n|^2 < \infty$ and the mean energy is finite. For $n = 6j+1$ the sine equals $\sqrt{3}/2$ and $n^4|c_n|^2 \to 81/\pi^2$. Therefore $\sum n^4|c_n|^2$ diverges and $\phi \notin D(H)$. The cut is at a zero of the original wavefunction, which removes the jump responsible for an infinite first energy moment at a nonnodal cut.

Exercise 3 (Inferring a relative phase). The initial box vector is $\psi = (u_1 + e^{\mathrm{i}\phi}u_2)/\sqrt{2}$, with an unknown $\phi \in \mathbb{R}$. Position means are measured on separately prepared copies at $t = 0$ and $t = \tau = \pi\hbar/(6E_1)$. They are

$$\langle Q\rangle_0 = \frac{L}{2} + \frac{8L}{9\pi^2}, \qquad \langle Q\rangle_\tau = \frac{L}{2} - \frac{8\sqrt{3}L}{9\pi^2}.$$

Find ϕ modulo 2π and describe the ambiguity if only the first mean is known. Use the given matrix element $\langle u_1, Qu_2\rangle = -16L/(9\pi^2)$.

Solution. Time evolution multiplies the second coefficient relative to the first by $e^{-\mathrm{i}3E_1 t/\hbar}$. The position matrix therefore gives

$$\langle Q\rangle_t = \frac{L}{2} - \frac{16L}{9\pi^2}\cos(\phi - 3E_1 t/\hbar).$$

At zero the measured mean gives $\cos\phi = -1/2$. At τ the cosine becomes $\sin\phi$, so the second mean gives $\sin\phi = \sqrt{3}/2$. Together these determine

$$\phi = \frac{2\pi}{3} \pmod{2\pi}.$$

The first mean alone permits both $2\pi/3$ and $4\pi/3$ modulo 2π. These are measurements on separately prepared copies, not two projective position readings on the same evolving particle.

Exercise 4 (A coarse energy reading keeps coherence). Let $\psi = (u_1+u_2+u_3)/\sqrt{3}$. Compare a yes/no reading of energy in $\{E_1, E_2\}$, conditioned on yes, with a fully resolved energy reading whose result is forgotten after retaining only E_1 or E_2. Give the two conditional density operators and their later position means. Use $\langle u_1, Qu_2\rangle = -16L/(9\pi^2)$.

Solution. The yes/no projection retains both amplitudes coherently. The two conditional density operators are

$$\rho_{\text{coarse}} = \frac{1}{2}|u_1+u_2\rangle\langle u_1+u_2|, \qquad \rho_{\text{resolved}} = \frac{1}{2}|u_1\rangle\langle u_1| + \frac{1}{2}|u_2\rangle\langle u_2|.$$

Each retention probability is $2/3$. The off-diagonal entries of the first operator evolve with the energy difference $3E_1$. The position means are

$$\langle Q\rangle_{\text{coarse},t} = \frac{L}{2} - \frac{16L}{9\pi^2}\cos\frac{3E_1 t}{\hbar}, \qquad \langle Q\rangle_{\text{resolved},t} = \frac{L}{2}.$$

Forgetting a resolved reading removes the cross term.